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Not All Collocation Errors Matter Equally: A Corpus-Based Study of Error Severity and Writing Quality in EFL Learners

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Abstract

This study investigates how collocation errors influence writing quality in EFL learners, with a particular focus on whether error frequency or error severity provides a more meaningful ex planation of writing performance. A corpus-based approach was adopted, in which 100 essays were randomly selected from approximately 10,000 entrance examination English essays. Col location errors were identified and categorized using the Taiwanese Corpus of English as a Foreign Language (TECCL). Multiple quantitative analyses—including frequency analysis, feature importance analysis, group comparison, and correlation analysis—were conducted to examine the relationship between collocation errors and writing scores. The results indicate that error frequency alone is not a reliable predictor of writing quality, as no consistent negative correlation was observed. In contrast, severity-weighted error measures demonstrate stronger explanatory power and clearer distinctions across different score groups. Furthermore, among various collocation subtypes, Verb–Noun and Adjective–Noun errors show a more significant impact on writing performance. Overall, the findings suggest that the influence of collocation errors on writing quality is not linear, but depends on both the severity and communicative im pact of errors, highlighting the importance of emphasizing collocation awareness and error se riousness in writing instruction.

Keywords: collocation errors, corpus-based analysis, EFL writing, error severity, writing quality

1 Introduction

With the increasing importance of English in global academic and international communication, English writing has become an essential skill for second language learners. In academic and standardized assessment contexts such as IELTS, writing reflects not only language proficiency but also the ability to organize and communicate ideas effectively. Since lexical resource is an important factor in writing quality (Abe, 2008), appropriate vocabulary use, especially natural word combinations, plays a crucial role in successful writing.

Collocations, or words that commonly occur together, are fundamental to natural and fluent language use. According to the Lexical Approach, language consists largely of common word combinations rather than isolated words (Lewis, 1993). However, EFL learners often struggle to use collocations accurately, and inappropriate collocation use may make writing sound unnatural and reduce overall quality (Mahmoud, 2005; Demir, 2017).

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Although many studies have examined collocation errors, most have focused on identifying error types and explaining their causes. Relatively few studies have investigated how collocation errors relate to writing performance, especially whether all errors influence writing quality equally. In particular, the role of error severity remains underexplored, raising the question of whether severe collocational misuse has a stronger impact than frequent but minor errors. This gap suggests the need to move beyond simple error counts and consider error type, severity, and writing scores together.

To address this gap, this study adopts a corpus-based quantitative approach to examine collocation errors in EFL learners' writing. Using 100 essays randomly selected from approximately 10,000 Chinese National College Entrance Examination English essays, the study combines corpus-based error identification with quantitative analysis in Orange Data Mining. It aims to determine which collocation-related features are most strongly associated with writing performance and to provide pedagogical implications for EFL writing instruction.

To address the research gap identified above, this study aims to answer the following research questions:
(1) What types of collocation errors are most frequently observed in EFL learners' writing?
(2) To what extent do collocation error frequency and severity influence writing quality?
(3) Which factor has a stronger relationship with writing performance: error type or error severity?
(4) Do different patterns of collocation errors distinguish higher-scoring and lower-scoring essays?

2 Literature Review

2.1 Second Language Writing and Standardized Writing Assessment

Second language (L2) writing is an important indicator of learners' language proficiency because it requires both linguistic knowledge and the ability to organize meaning effectively. In academic and examination contexts, writing is used to evaluate learners' ability to express ideas clearly, use vocabulary appropriately, and construct coherent texts. Vocabulary use is especially important in standardized writing assessments such as IELTS, where lexical resource reflects the appropriateness, range, and naturalness of word use (Abe, 2008). Since collocations are closely related to natural and appropriate vocabulary use, they are relevant to the evaluation of EFL writing performance.

2.2 Collocational Competence and Writing Quality

Collocations are an essential part of vocabulary knowledge because they help language users produce natural and fluent expressions. Lewis (1993) argues that language consists largely of lexical chunks and common word combinations rather than isolated words, while Mahmoud (2005) emphasizes that appropriate collocation use contributes to the naturalness and acceptability of written language. In L2 writing, collocational competence can affect how readers perceive text quality because unnatural word combinations may weaken clarity, fluency, or lexical appropriateness, even when the grammar is correct (Demir, 2017).

However, previous research has not fully explained which aspects of collocation use matter most for writing quality. Some errors may occur frequently but have limited influence on meaning, whereas others may occur less often but seriously affect clarity or naturalness. This distinction

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suggests that collocation errors should not be evaluated only by frequency, but also by their severity and communicative effect.

2.3 Collocation Errors in EFL Writing

Collocation errors are common in EFL writing because learners often find it difficult to produce natural English word combinations. Mahmoud (2005) notes that such errors may result from direct translation from learners' first language, leading to expressions that are grammatically possible but unnatural or inappropriate. These errors differ from purely grammatical errors because they are closely related to lexical appropriateness, semantic clarity, and naturalness.

Collocation errors can be classified according to word-combination patterns. Benson et al. (1986) identify categories such as verb–noun, adjective–noun, noun–noun, adverb–adjective, and preposition-related combinations. This classification makes it possible to examine error patterns systematically and determine whether certain subtypes are more closely related to writing quality. However, error type alone may not fully explain writing performance, because errors within the same category may differ in seriousness. Therefore, both error type and severity should be considered.

2.4 Error Severity and Writing Evaluation

Error severity is important in writing evaluation because not all errors affect writing quality to the same degree. Some errors are minor and do not prevent readers from understanding the intended meaning, while others reduce clarity, naturalness, or communicative effectiveness. This distinction is especially relevant to collocation errors. A learner may produce many minor collocation errors that only make the text slightly unnatural, while another may produce fewer but more serious errors that damage clarity. Therefore, frequency counts alone may overlook important differences in error impact.

In this study, error severity is used as an analytical dimension to complement error frequency and error type. By assigning different weights to minor, moderate, and severe collocation errors, the study captures the relative impact of errors on meaning, appropriateness, and reader comprehension.

2.5 Corpus-Based Approaches to Collocation Analysis

Corpus-based approaches provide a systematic way to analyze language use because they allow researchers to examine patterns across authentic texts. Biber et al. (1998) argue that corpus linguistics offers empirical evidence for studying language structure and use. In collocation research, corpus-based methods are useful because collocations are based on patterns of word co-occurrence, making it possible to identify whether a learner's word combination is natural, uncommon, or inappropriate.

Learner corpora are also valuable for studying EFL writing because they contain real examples of learner language. Granger (1998) emphasizes their importance for understanding learner errors and language development. In this study, the Taiwanese Corpus of English as a Foreign Language (TECCL) is used as a reference for identifying learner collocation patterns. Corpus-based analysis is also combined with quantitative measures, including writing scores, error frequency, and error severity, to examine how collocation errors relate to writing performance.

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2.6 Research Gap

Previous studies have shown that collocational competence is important for natural and effective EFL writing and have identified common types and causes of collocation errors. However, fewer studies have examined how different collocation errors are associated with writing scores. In particular, the distinction between error frequency and error severity remains insufficiently explored.

This gap is important because frequent errors do not necessarily have the strongest effect on writing quality. Some frequent errors may have limited influence on meaning, while less frequent but more severe errors may affect clarity, naturalness, and communicative effectiveness more directly. To address this gap, the present study adopts a corpus-based quantitative approach that integrates collocation error classification, severity weighting, and writing-score analysis.

3 Methodology

This study adopts a corpus-based quantitative research design to examine the relationship between collocation errors and writing quality in EFL learners' essays. The analysis followed five main stages: data sampling, collocation error identification, error classification, severity coding, feature construction, and statistical analysis. This design allows the study to compare frequency-based and severity-based measures rather than assuming that all collocation errors have the same impact on writing quality.

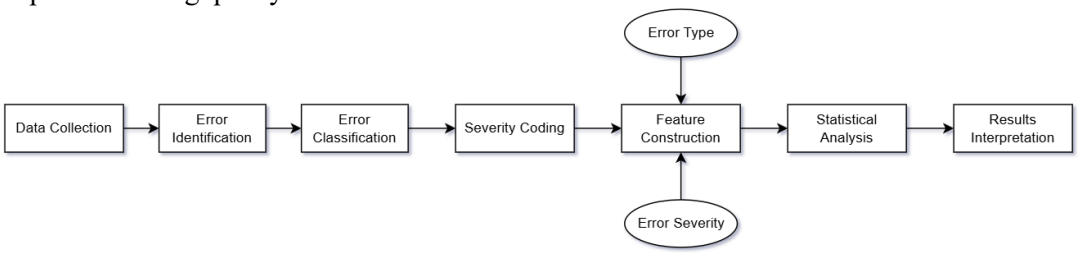


Figure 1. Framework of the Research Design

Figure 1 presents the overall research procedure, from essay sampling to statistical analysis. The procedure was designed to move beyond simple error counting by comparing total error frequency, subtype frequency, and severity-weighted error scores in relation to writing scores.

3.1 Data Source and Sampling

The data were drawn from English writing essays collected from the Chinese National College Entrance Examination. The original dataset contains approximately 10,000 essays written by EFL learners. Because detailed manual identification and classification of collocation errors were required, 100 essays were randomly selected for close analysis.

To ensure consistency, the selected essays had to be based on the same or similar writing task, be complete enough for linguistic analysis, and contain sufficient sentence-level writing for collocation errors to be identified. Incomplete, extremely short, or off-topic essays were excluded. The writing score of each essay was used as the indicator of writing quality and served as the dependent variable in the quantitative analysis.

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3.2 Collocation Error Identification

Collocation errors were identified by examining word combinations in each essay. In this study, a collocation error refers to a word combination that is grammatically possible but unnatural, inappropriate, uncommon, or unclear in context. The focus was placed on lexical combinations rather than general grammatical mistakes. For example, strong rain was treated as a collocation error because heavy rain is the more natural expression.

The identification process involved manual reading, checking potential errors against corpus-based evidence and reference materials, including TECCL and English dictionaries, and evaluating each expression in context. This procedure helped distinguish collocation errors from grammatical errors, such as subject–verb agreement mistakes.

3.3 Error Classification

After identification, collocation errors were classified according to common word-combination patterns, including Verb–Noun, Adjective–Noun, Noun–Noun, Adverb–Adjective, Preposition-related, and Other lexical collocations. This classification provided a systematic basis for comparing whether certain subtypes were more strongly associated with writing performance than others.

Error Type	Description	Example
Verb–Noun	An unnatural combination of a verb and a noun	*make a homework → do homework*
Adjective–Noun	An inappropriate adjective used with a noun	*strong rain → heavy rain*
Noun–Noun	An unnatural noun combination	*knowledge learning → knowledge acquisition*
Adverb–Adjective	An inappropriate adverb used with an adjective	*very excellent → excellent*
Preposition-related	An incorrect or unnatural preposition pattern	*depend of → depend on*
Other lexical collocation	A collocation error that did not fit the above categories	context-dependent

Table 1. Classification of Collocation Error Types

This classification allowed the study to compare whether certain collocation subtypes were more strongly associated with writing performance than others.

3.4 Error Severity Coding

Each collocation error was also coded according to severity because not all errors influence writing quality to the same degree. A three-level scale was used: minor, moderate, and severe. Each level was assigned a numerical weight for quantitative analysis.

Severity Level	Weight	Definition	Example
Minor	1	The meaning is clear, but the expression is slightly unnatural or less idiomatic.	*do a mistake → make a mistake*
Moderate	2	The meaning is understandable, but the expression is noticeably unnatural or weakens clarity.	*strong rain → heavy rain*

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Severe	3	The error causes unclear, misleading, or seriously inappropriate meaning.	*context-dependent expressions that interfere with comprehension*

Table 2. Severity Coding Scheme for Collocation Errors

The weighted collocation error score was calculated by multiplying each error by its severity weight and summing the total for each essay. The formula was:

Weighted Collocation Error Score = Number of Minor Errors × 1 + Number of Moderate Errors × 2 + Number of Severe Errors × 3

The weighted score was used to represent the overall severity of collocation errors in each essay.

3.5 Feature Construction

After error identification and severity coding, analytical features were constructed for each essay. These features included total error frequency, subtype frequency, severity counts, and weighted collocation error scores. The writing score was used as the dependent variable representing writing quality.

Feature	Description
Total collocation error frequency	Total number of collocation errors in each essay
Error subtype frequency	Number of errors in each collocation category
Minor error count	Number of minor collocation errors
Moderate error count	Number of moderate collocation errors
Severe error count	Number of severe collocation errors
Weighted collocation error score	Total severity-weighted score for each essay
Writing score	Original essay score used as the writing quality indicator

Table 3. Analytical Features Used in the Study

These features made it possible to compare frequency-based and severity-based explanations of writing quality.

3.6 Data Analysis

The data were analyzed using descriptive statistics, group comparison, correlation analysis, and feature importance analysis. Descriptive statistics were used to identify the frequency and proportion of different collocation error types. Group comparison examined whether essays at different score levels showed different patterns of error frequency and severity. Correlation analysis tested the relationship between writing scores and both frequency-based and severity-based error measures. Finally, feature importance analysis using the RReliefF algorithm was conducted to identify which error features contributed most strongly to writing score differences.

3.7 Analysis Tool

Orange Data Mining was used as the main tool for data analysis and visualization. The dataset was organized in spreadsheet format, with each row representing one essay and each column representing one analytical feature. Orange was used to conduct descriptive statistics, correlation analysis, visualization, and RReliefF feature importance analysis.

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4 Results

4.1 Relationship Between Collocation Error Frequency and Writing Scores

Figure 2 presents the relationship between the total frequency of collocation errors and writing scores. The scatter plot shows that the data points are widely distributed across different score levels. No clear downward trend can be observed between the number of collocation errors and writing scores. This indicates that essays with more collocation errors did not necessarily receive lower scores, while some essays with fewer collocation errors did not always receive higher scores.

This result suggests that simple error frequency alone may not be sufficient to explain differences in writing quality. In other words, counting the total number of collocation errors provides only a limited view of how collocation use relates to writing performance.

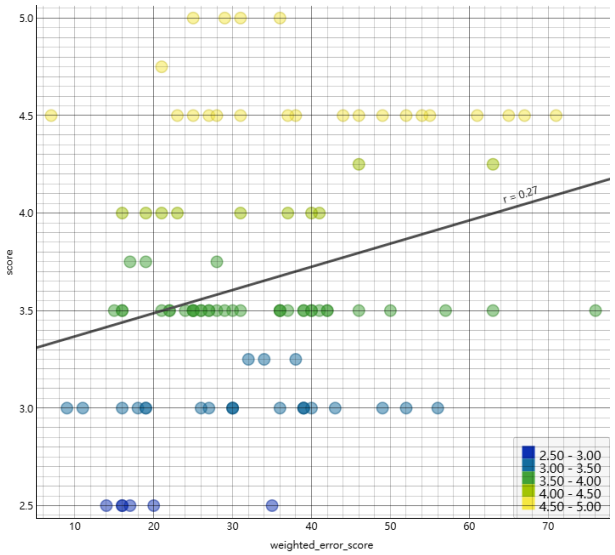


Figure 2. Relationship Between Collocation Error Frequency and Writing Score

4.2 Frequency Distribution of Collocation Error Subtypes

Figure 3 shows the frequency distribution of different collocation error subtypes. Among the identified error categories, Verb–Noun errors occurred most frequently, followed by preposition-related errors and noun-related collocation errors. Other subtypes, such as Adjective–Noun and Adverb–Adjective errors, appeared less frequently in the dataset.

This distribution indicates that learners had particular difficulty with core lexical combinations, especially Verb–Noun patterns. However, the frequency of an error subtype does not necessarily mean that it has the strongest influence on writing scores. Therefore, further analysis was conducted to examine the relationship between error severity, error type, and writing performance.

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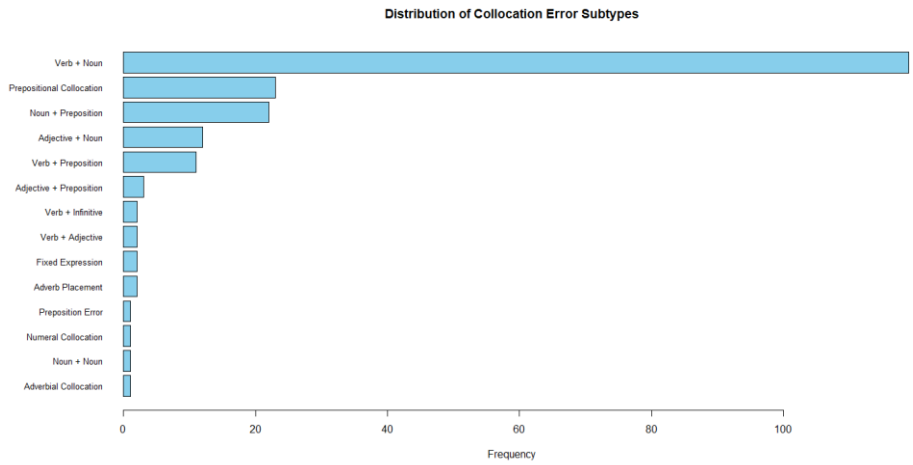


Figure 3. Frequency Distribution of Collocation Error Subtypes

4.3 Error Severity and Writing Performance

Figure 4 presents the distribution of severity-weighted collocation error scores across writing levels. Compared with simple error frequency, the weighted error score shows a clearer distinction between score groups. Lower-scoring essays tended to have higher weighted error scores, while medium-scoring essays showed relatively lower weighted error scores.

This pattern suggests that the severity of collocation errors may provide a more informative measure than frequency alone. While the total number of errors did not show a consistent relationship with writing scores, severity-weighted errors appeared to better reflect differences in writing performance.

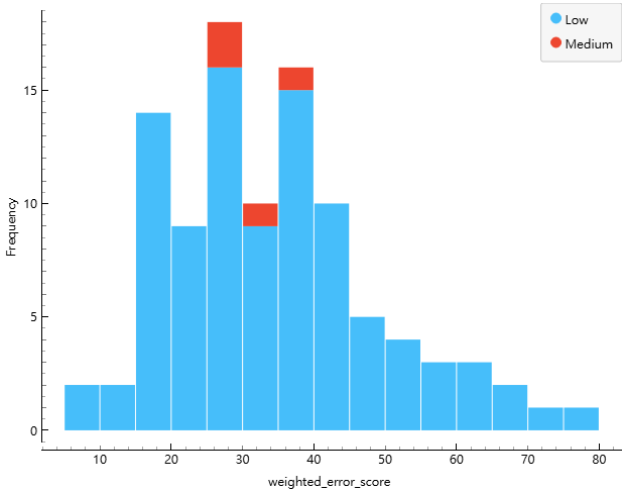


Figure 4. Severity-Weighted Collocation Error Scores Across Writing Levels

4.4 Relative Contribution of Collocation Error Subtypes

Figure 5 compares the relative contribution of different collocation error subtypes to writing scores. The results show that Verb–Noun errors had the largest relative contribution among the

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collocation subtypes, followed by Adjective–Noun errors. Other subtypes showed relatively smaller associations with writing scores.

This result suggests that not all collocation error types are equally related to writing performance. Errors involving central lexical combinations, such as Verb–Noun and Adjective–Noun patterns, appeared to be more closely associated with writing scores than less central or less frequent collocation patterns.

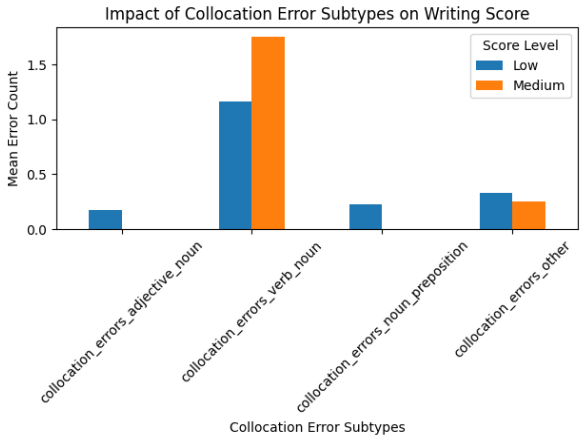


Figure 5. Relative Contribution of Collocation Error Subtypes to Writing Scores

4.5 Correlation Between Error Measures and Writing Scores

Figure 6 compares the correlation between different error measures and writing scores. The results show that severity-based collocation error measures had a stronger association with writing scores than frequency-based measures. In contrast, grammatical error measures showed weaker correlations with writing scores.

This comparison supports the view that the impact of errors on writing quality cannot be fully explained by frequency. Although grammatical errors may occur frequently, they did not show as strong a relationship with writing scores as collocation-related measures. This finding indicates that error type and error severity should both be considered when analyzing writing quality.



Figure 6. Correlation Between Error Measures and Writing Scores

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4.6 Comparison Between Error Frequency and Error Impact

Figure 7 compares the average frequency of grammatical errors and collocation errors with their correlation with writing scores. The results show that grammatical errors occurred more frequently than collocation errors. However, collocation errors showed a stronger association with writing scores than grammatical errors.

This contrast indicates that the most frequent error type is not necessarily the most influential one. Although grammatical errors were more common in the essays, collocation errors appeared to be more closely related to writing performance. Therefore, the results suggest that writing quality is influenced not only by how many errors learners make, but also by what types of errors they make and how strongly those errors affect meaning and naturalness.

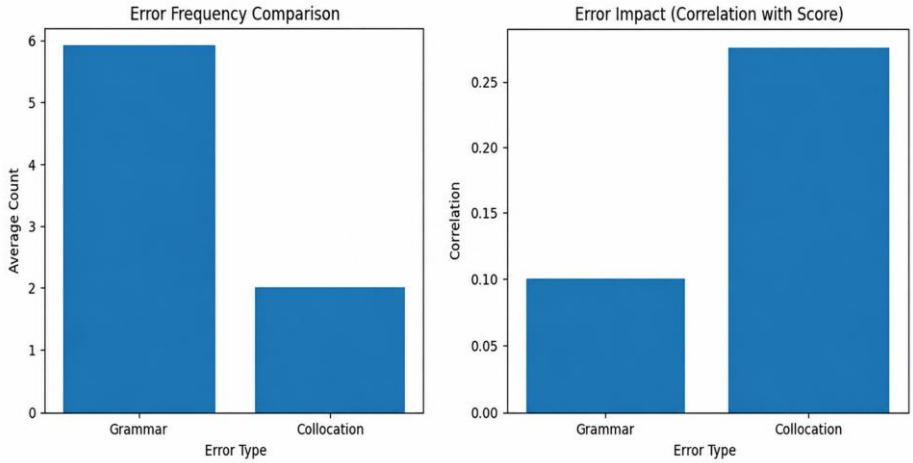


Figure 7. Comparison of Error Frequency and Correlation with Writing Scores

5 Discussion

The findings of this study provide insights into how collocation errors relate to EFL writing quality. The total frequency of collocation errors did not have a clear negative relationship with writing scores, suggesting that simply counting errors is not sufficient to explain writing performance. In contrast, severity-weighted error scores showed clearer differences across writing levels, indicating that the seriousness of collocation errors may provide a more meaningful explanation of writing quality. Minor errors may only make expressions slightly unnatural, whereas severe errors may reduce clarity, precision, or communicative effectiveness.

The results also indicate that different collocation subtypes are not equally related to writing performance. Verb–Noun and Adjective–Noun errors appeared to have stronger associations with writing scores than other subtypes, possibly because these combinations often carry central semantic information. When learners use unnatural verb–noun or adjective–noun combinations, the sentence may become less precise or less natural even if it remains grammatically understandable.

These findings extend previous research on collocations in EFL writing by showing that the influence of collocation errors is not uniform. While previous studies have emphasized the importance of collocational competence, the present study further suggests that collocation errors

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should be examined through both error type and error severity. In this sense, collocation errors should not be treated as simple vocabulary mistakes, but as lexical problems that may affect semantic naturalness, clarity, and communicative effectiveness.

Pedagogically, the findings suggest that EFL writing instruction should place greater emphasis on collocation awareness. Instead of focusing mainly on grammar correction, teachers may help learners notice common collocation patterns, especially Verb–Noun and Adjective–Noun combinations. Feedback should also prioritize more serious collocation errors that interfere with meaning or clarity, rather than treating all errors equally.

This study has several limitations. First, the sample size was limited to 100 essays, which may restrict the generalizability of the findings. Second, collocation error identification and severity coding involved some judgment, although corpus-based references and dictionaries were used. Third, the study focused mainly on collocation errors and did not fully examine other aspects of writing quality, such as coherence, organization, and task response. Future research could use a larger dataset, involve multiple raters, and examine how collocation errors interact with other linguistic and discourse-level features.

Overall, the findings suggest that collocation errors should not be treated as equal in writing analysis or instruction. Their influence depends not only on how often they occur, but also on their severity and subtype. Therefore, a more meaningful understanding of EFL writing quality requires attention to how errors affect naturalness, clarity, and communication.

6 Conclusion

This study investigated how collocation errors relate to writing quality in EFL learners' essays. By adopting a corpus-based quantitative approach, the study examined collocation error frequency, error type, and error severity in relation to writing scores. The findings suggest that error frequency alone does not provide a sufficient explanation of writing performance. Instead, severity-weighted collocation error scores appeared to offer a clearer indication of differences across writing levels.

The results also indicate that different collocation error subtypes may have different levels of influence on writing quality. In particular, Verb–Noun and Adjective–Noun errors appeared to be more closely associated with writing scores than other error types. This suggests that collocation errors should not be treated equally, as some errors may affect naturalness, clarity, and communicative effectiveness more strongly than others.

From a pedagogical perspective, the findings highlight the need to place greater emphasis on collocation awareness in EFL writing instruction. Teachers may help learners improve writing quality by focusing not only on grammatical accuracy, but also on natural and appropriate word combinations. Corpus-based resources may also support learners in noticing common collocation patterns and reducing serious lexical errors.

This study is limited by its relatively small sample size and by the judgment involved in identifying and coding collocation errors. Future research could use a larger dataset, involve multiple raters, and examine the relationship between collocation errors and other aspects of writing quality, such as coherence and organization. Overall, the study suggests that not all collocation errors

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matter equally, and that error severity and error type should be considered when analyzing EFL writing performance.

7 References

[1] M. Abe, "Exploring the Role of Model Essays in the IELTS Writing Test: A Feedback Tool," Master's thesis, The University of Queensland, 2008. Available: https://espace.library.uq.edu.au/

[2] M. Benson, E. Benson, and R. Ilson, *The BBI Combinatory Dictionary of English: A Guide to Word Combinations*. Amsterdam, Netherlands: John Benjamins, 1986.

[3] D. Biber, S. Conrad, and R. Reppen, *Corpus Linguistics: Investigating Language Structure and Use*. Cambridge, U.K.: Cambridge University Press, 1998.

[4] P. Cantos-Gómez and M. Almela-Sánchez, Eds., *Lexical Collocation Analysis: Advances and Applications*. Cham, Switzerland: Springer, 2018, doi: 10.1007/978-3-319-92582-0.

[5] J. W. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 4th ed. Thousand Oaks, CA, USA: Sage, 2014.

[6] C. Demir, "Lexical Collocations in English: A Comparative Study of Native and Non-Native Scholars of English," *Journal of Language and Linguistic Studies*, vol. 13, no. 1, pp. 75–87, 2017. Available: https://www.jlls.org/index.php/jlls/article/view/738

[7] Z. Dörnyei, *Research Methods in Applied Linguistics*. Oxford, U.K.: Oxford University Press, 2007.

[8] S. Granger, Ed., *Learner English on Computer*. London, U.K.: Longman, 1998.

[9] M. Lewis, *The Lexical Approach: The State of ELT and a Way Forward*. Hove, U.K.: Language Teaching Publications, 1993.

[10] A. Mahmoud, "Collocation Errors Made by Arab Learners of English," *Asian EFL Journal*, vol. 5, pp. 117–126, 2005. Available: https://www.asian-efl-journal.com/

[11] L. J. Zhang, "Second Language Writing as and for Second Language Learning," *Journal of Second Language Writing*, vol. 22, no. 4, pp. 446–447, 2013, doi: 10.1016/j.jslw.2013.08.010.

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